Prediction of protein regions susceptible to proteolytic processing

Eugene V. Matveev[1], Vyacheslav V. Safronov[2], Marat D. Kazanov[1,3]

Correspondence to: m.kazanov@skoltech.ru

[1] Institute for Information Transmission Problems, Russia;
[2] Faculty of bioengineering and bioinformatics, Lomonosov Moscow State University, Russia
[3] Skolkovo Institute of Science and Technology, Russia

Introduction

Regulatory proteolysis or proteolytic processing, which is the main topic of this study, is a process of activation or deactivation of a target protein via hydrolysis of its one or few peptide bonds by a specialized enzyme, protease. Contrary to proteases involved in a food digestion, regulatory proteases usually possess high specificity, which allows selectively bind its substrate for the modulation of its function. Proteases recognize their cleavage sites primarily based on a specific amino acid context around the cleaved peptide bond. Stretches of amino acid sequences satisfying specificity of regulatory proteases are rarely appears at random and if occurs, especially if they are also accessible in a structural sense, in most cases represent native cleavage sites. Thus, a prediction of cleavage sites of regulatory proteases based on the known specificity of protease and known 3D structure of substrate can be a powerful approach for the identification of protease substrates. In this study we developed the computational model predicting structural susceptibility of proteins to proteolytic processing.

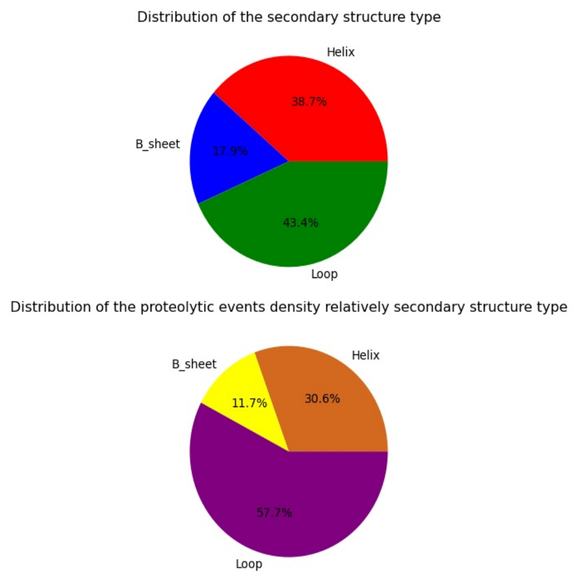
Methods

We developed a new predictive model based on experimentally registered proteolytic events obtained from the CutDB database. The whole approach includes developing of the separate predictive models for three different types of secondary structure and application of positive-unlabeled (PU) machine learning approaches due to presence of hidden (unregistered) cleavage sites in the training dataset. At first, we performed a mapping of the CutDB proteolytic events on 3D protein structures obtained from PDB database. Then, we calculated structural features using structural bioinformatics tools (DSSP, etc.) and introduced several ad-hoc structural features. Finally, we applied different machine learning methods, estimated the quality of the obtained models and applied the PU-learning algorithm for correcting probabilities of predicted susceptibilities of protein regions to proteolytic processing.

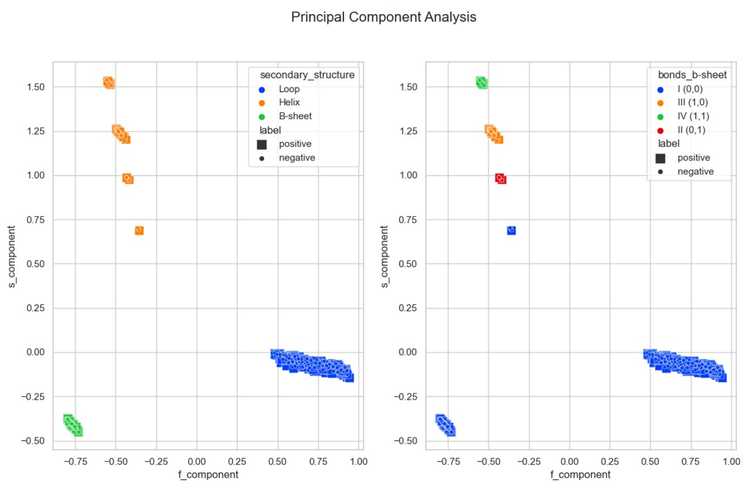
Results

Training set description

Num rows in dataset:	88219
Num of proteolytic events:	444
% of proteolytic events:	0.503%
Helix	
num of aa in helix type of SS:	34146
Num of proteolytic events in helix type of SS	136
B-sheet	
num of aa in b-sheet type of SS	15791
num of proteolytic events in b-sheet type of SS	52
Loop	
num of aa in loop type of SS	38282
num of proteolytic events in loop type of SS	256

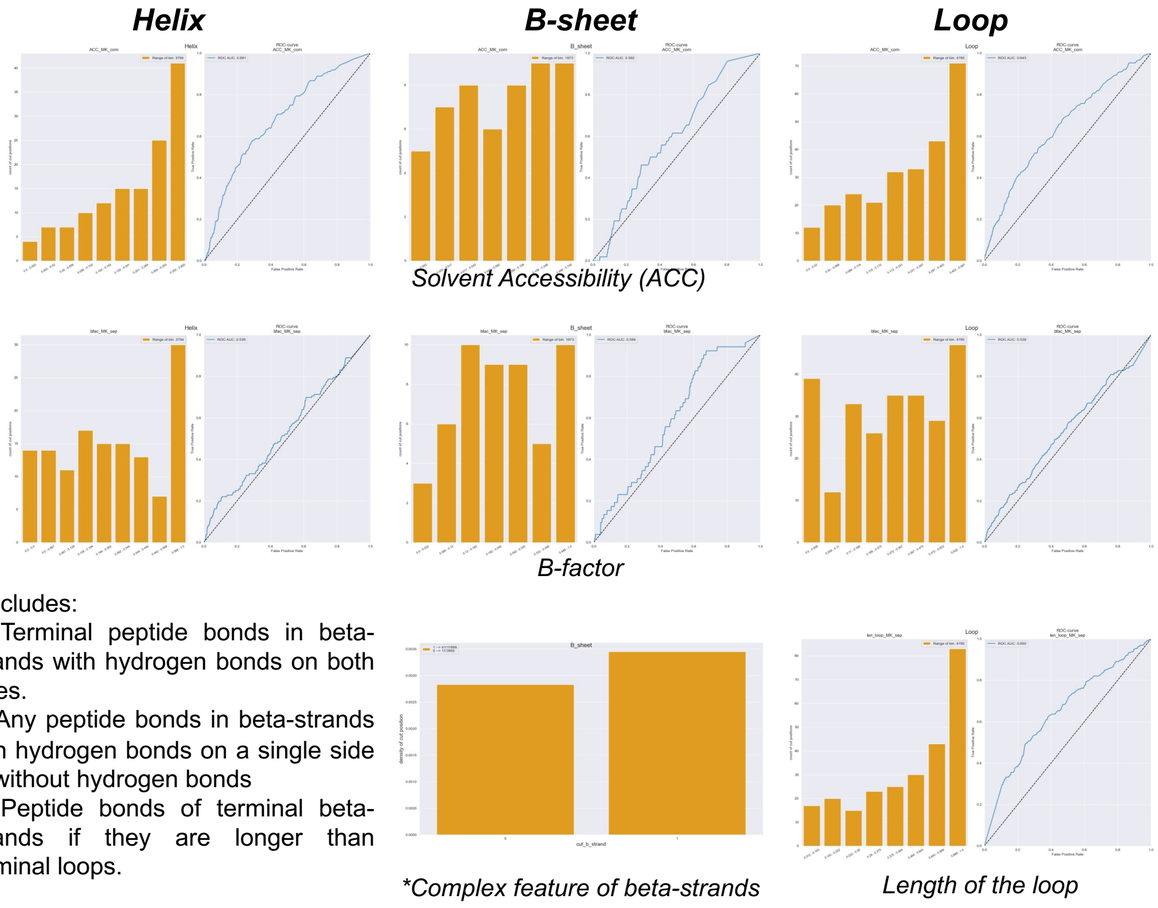


Structural feature analysis



Different clusters of peptide bonds visualized by PCA method led us to the idea of constructing separate predictive models for each type of secondary structure and generalizing it into a single predictive model.

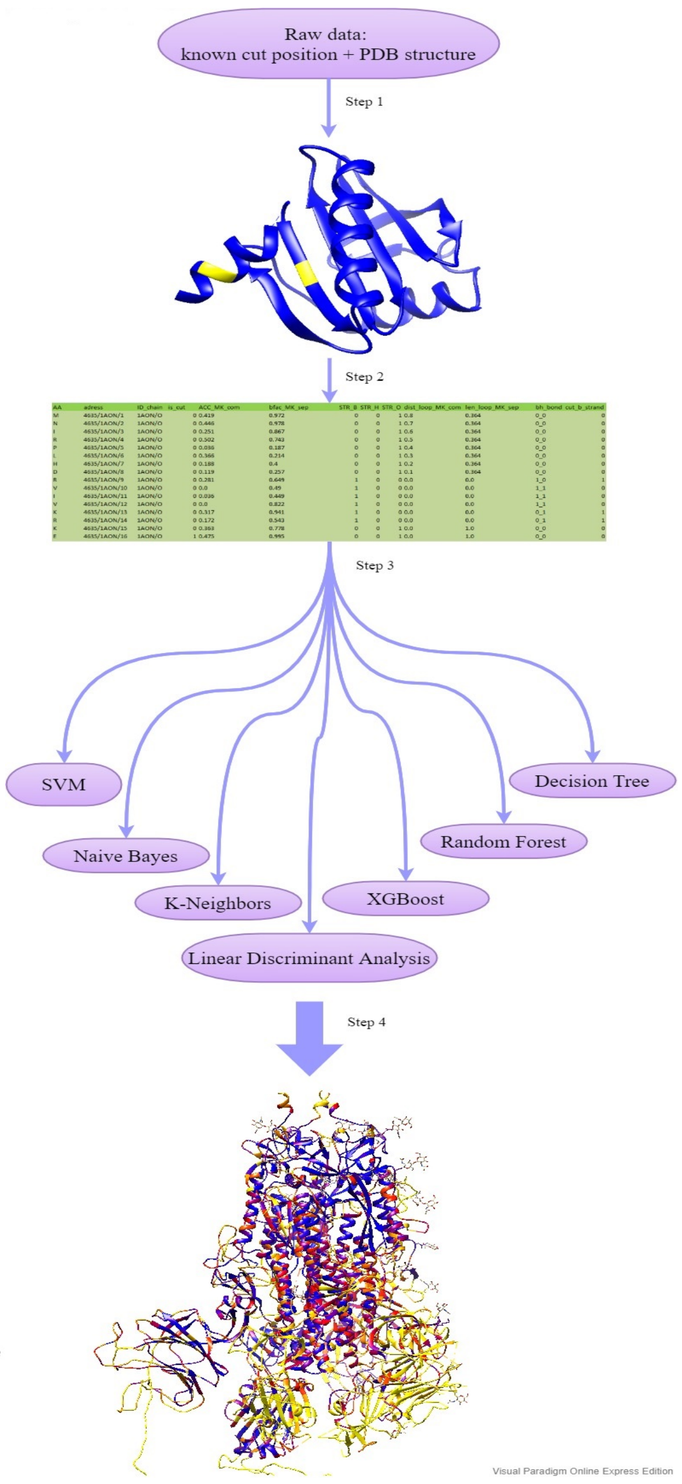
Analysis of predictive capabilities of structural features



* Includes:
1) Terminal peptide bonds in beta-strands with hydrogen bonds on both sides.
2) Any peptide bonds in beta-strands with hydrogen bonds on a single side or without hydrogen bonds
3) Peptide bonds of terminal beta-strands if they are longer than terminal loops.

Acknowledgements

We are thankful to Andrei Osterman and Jeff Smith lab members (Sanford-Burnham Medical Research Institute, La Jolla, USA) for useful discussions. This reported study was funded by RFBR according to the research project #20-04-60066.



Development workflow

Step 1

Based on known sequences of protease substrates obtained from the CutDB database we selected from PDB database homologous 3D structures having > 90% sequence identity. We visualized cut sites by mapping to the obtained 3D structures and curated the whole dataset.

Step 2

a) Calculating structural features using structural bioinformatics tools (DSSP, etc.) and creation of ad-hoc structural features.
b) Feature transformation with one-hot encoding technique for categorical variable (type of secondary structure).
c) Feature normalization and standardization for continuous variables.
d) Data curation and feature selection.

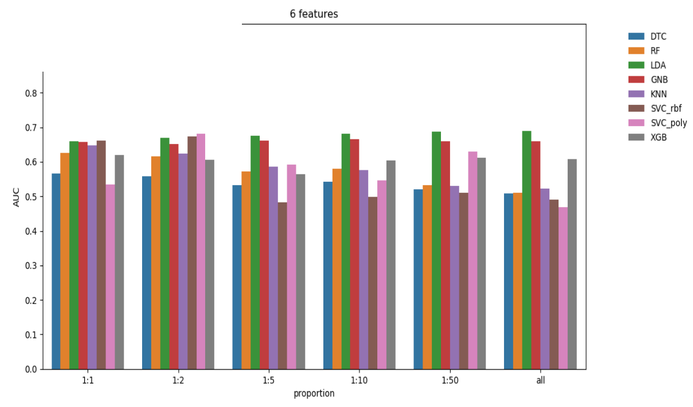
Step 3

Application of various machine learning method to several versions of training dataset having different ratios of positive/unlabeled class sizes. We used training datasets with the 1:1, 1:2, 1:5, 1:10, 1:50 positive/unlabeled class size ratios including the whole training dataset. Predictive quality of the constructed models was estimated by Area Under ROC curve (AUC).

Step 4

Application of the PU-learning approaches for correcting predictive probabilities and visualization of the calculating probabilities on 3D structures of potential substrates.

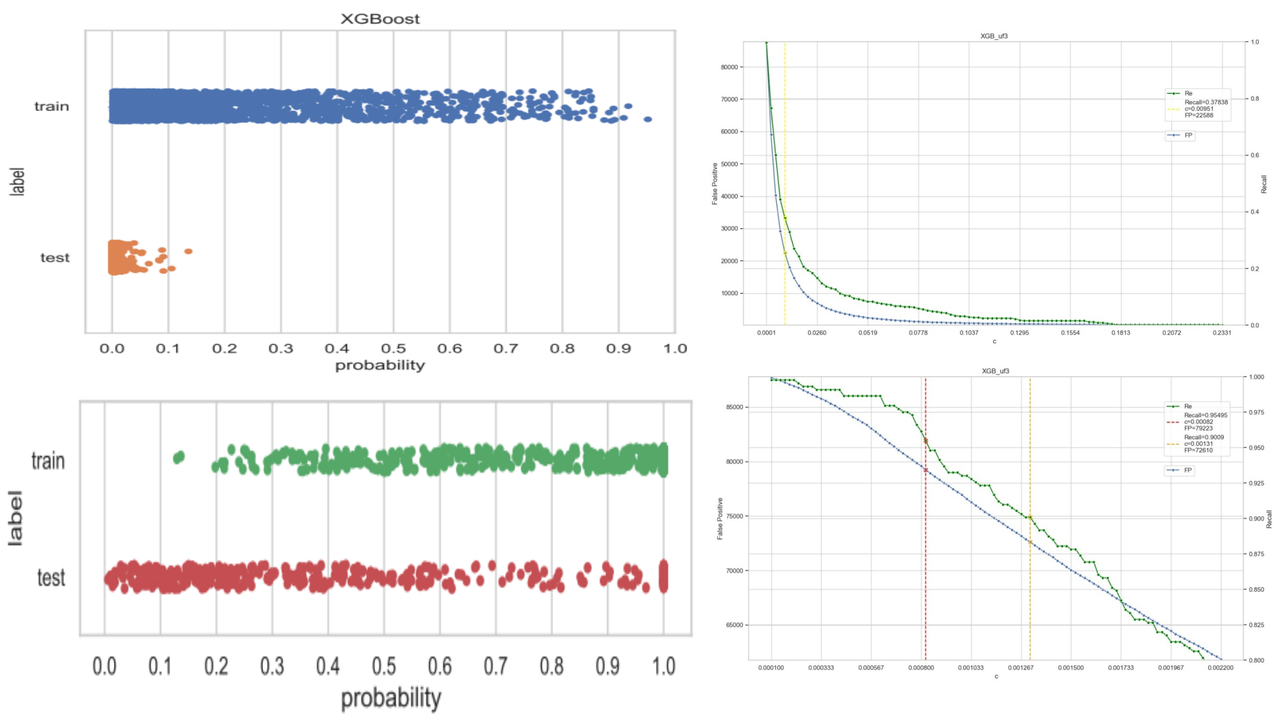
Selection of appropriate machine learning method



*All applied ML methods are from *scikit-learn* library except for XGBoost, which is from *xgboost* library.
Notations:
DTC – DecisionTreeClassifier
RF – RandomForestClassifier
LDA – LinearDiscriminantAnalysis
GNB – GaussianNB
KNN – KNeighborsClassifier
SVC_rbf – SVC with *rbf* kernel
SVC_poly – SVC with *polynomial* kernel
XGB - XGBClassifier

Correction of the predicted probabilities by PU-learning approach

Due to inevitable presence of the unregistered proteolytic events in the unlabeled part of the training dataset the prediction probabilities can be lower than it should be and need a correction. To this end, we applied the Positive-Unlabeled (PU) learning approach (Elkan and Noto method) adjusting probabilities by a coefficient C. Plots on the left part of the figure demonstrated the distribution of predicted probabilities before (upper plot) and after (lower plot) the correction. Coefficient C was found minimizing the number of False Positives (right side plots) preserving the high value of Recall (more than 90%).



Conclusion

Using CutDB data on registered proteolytic events we studied prediction power of the structural features of potential protease substrates based on information on its 3D structure. The nature of the dataset with hidden objects of Positive class makes suitable the application of the positive-unlabeled ML approaches. Prediction power of structural features varies with the secondary structure implying the utility of making separate models for each type of secondary structure.